Chapter 1

The Greek Alphabet

If you do anything involving math or physics, you will be encountering Greek letters on a regular basis. Here is a table for your reference:

Capital	Lower	Pronounced	Capital	Lower	Pronounced
A	α	Alpha	N	ν	Nu
B	β	Beta	Ξ	ξ	Xi ("ku-ZY")
Γ	γ	Gamma	O	o	Omicron
Δ	δ	Delta	Π	π	Pi
E	ϵ	Epsilon	P	ρ	Rho
Z	ζ	Zeta	Σ	σ	Sigma
H	η	Eta	T	τ	Tau
Θ	θ	Theta	Υ	υ	Upsilon
I	ι	Iota	Φ	ϕ	Phi
K	κ	Kappa	X	χ	Chi ("Kai")
Λ	λ	Lambda	Ψ	ψ	Psi ("Sigh")
M	μ	Mu	Ω	ω	Omega

1.1 Greek Letters in Math and Science

These are important to know, for a large variety of reasons. Let's look at the most common physics and math uses:

Symbol	Common meaning
π	circle constant, approximately 3.14159
θ	an angle
Δ	change in a quantity
Σ	sum of many terms
λ	wavelength
μ	coefficient of friction, mean, or micro-
ρ	density
ω	angular velocity
Ω	ohms, a unit of electrical resistance

You will see a lot of equations that reference Greek letters, using them as *placeholders*, *constants*, or *variables* for an unknown quantity. For example, there area of a circle is given by

$$A = \pi r^2$$

where r is the radius of the circle. In this case, π is a constant, and r is a variable. In other cases, you might see the physics equation

$$\Delta x = v\Delta t + \frac{1}{2}a\Delta t^2$$

This is the equation for the change in position under constant acceleration, where Δx is the change in position, v is the initial velocity, a is the acceleration, and Δt is the change in time. In this case, Δx, v, a, and Δt are all variables.

One final example that we will explore later is the equation for wavelength in terms of frequency and the speed of light:

$$\lambda = \frac{c}{f}$$

where λ is the wavelength, c is the speed of light, and f is the frequency. In this case, λ and f are variables, while c is a constant. We can then translate lambda into other equations as needed.

Generally, we will define the Greek letters when providing an equation. However, it is important to be familiar with the Greek alphabet, so that you can understand what the symbols mean when you see them in equations.

Appendix A

Answers to Exercises